

1 **Design & Implementation of Human-Computer Interfaces** **NPTEL-MOOCs**

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2 **L7 – Functional Specification of Requirements**

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3 **Recap - ISDLC**

4 **Recap**

- We learned about
 - The idea of usability
 - Need for usability requirement gathering
 - Contextual inquiry – one such technique

5 **Recap**

- In this lecture – we shall learn about notations and conventions for system requirement specification

6

7 **Requirements**

- Earlier, we have discussed ONE type of requirements
 - Called “non-functional requirements” (NFR)

8 **Requirements**

- We have discussed different NFRs

9 **Broad Categories**

- Performance related
- Operating constraints
- Economic considerations
- Life-cycle requirements
- Interface issues

10 **Requirements**

- Usability requirements is one type of NFR
 - Very important for interactive system design

11 **Requirements**

- NFRs important
 - However, they are difficult to formally specify for implementation

12 **Requirements**

- System requirements should be “easy” to understand and implement (as code)
- Requires some formal way to specify

13 **Functional Requirements**

- Such system requirements, which can be formally specified, are generally termed as “functional requirements” (FR)
 - They are easy to convert into codes

14 **FR**

- As the name suggests, FR specifies a system in terms of “functions” – including input and output (and also purpose)

15 **Functional Requirement Specification**

16 **FR - Basics**

- Main idea
 - Abstraction - identify black box functions to be supported by the system
 - With clearly defined input and output

17 **FR - Basics**

- Main idea
 - Decomposition & hierarchy - to better manage the functional description

18 **FR - Functions**

- Key points to remember while identifying functions
 - Avoid exploratory style – from implementation to function (it should be other way round)
 - Abstraction
 - Hierarchy

19 **SRS**

- When we put requirements in textual form, we get the SRS (Software Requirement Specification) document
 - The “formal” representation/specification of requirements

20 **SRS**

- Hierarchical structure (R1, R1, R1.1 etc)
- Each R (e.g., R1, R1.1 etc) represents a function with clearly defined input and output and some description (optional)

21

Example

22 **An Example**

- Functional requirements of a Class-note Taking Application, focusing mainly on the needs of college-going students

23 **(Some) Requirements (Functional)**

- Requirement for “authentication” – for a user to access the app features
 - Requirement to “create account”
 - Requirement to “log in”
 - Requirement for “account management”

24 **A Hierarchical Depiction**25 **(Some) Requirements (Functional)**

- Requirement for “managing” the notebook
 - Requirement to “create/delete note”
 - Requirement to “open/close note”
 - Requirement for “edit note”
 - Requirement for “view all the notes”

26 **A Hierarchical Depiction**27 **(Some) Requirements (Functional)**

- Requirement for “searching” in the notebook
 - Requirement to “search by text”
 - Requirement to “search by date”

28 **A Hierarchical Depiction**29 **Complete Hierarchy**30 **Converting to SRS (Authentication)**31 **Converting to SRS (Authentication)**32 **Converting to SRS (Authentication)**33 **Converting to SRS (Authentication)**34 **Converting to SRS (Authentication)**

35 **Converting to SRS (Authentication)**

36 **Note**

- How functions are represented (name, input, output, description)
- Abstract (black-box) nature of the functions – what we need, NOT how to get it!
- Hierarchical organization
- Naming convention

37 **Take Home Exercise**

- Complete the SRS!

38 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 4
- Roger S Pressman (2020). Software Engineering: A Practitioner's Approach, 9th ed, McGraw-Hill Education, New York, Chapter 7-8